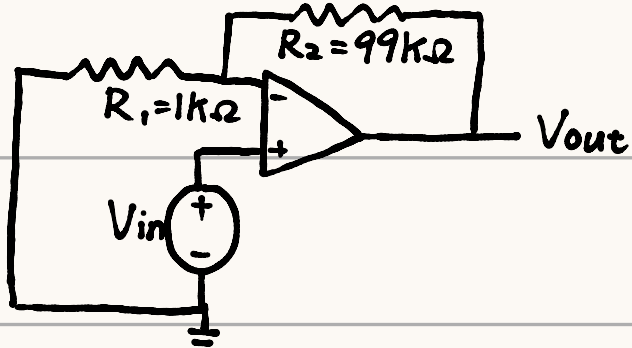


HW4

1. $0 \sim 50^\circ\text{C}$ $1\text{m}/^\circ\text{C} \Rightarrow V_{\text{in}} \in [0, 50\text{mV}]$

$$V_{\text{out max}} = 5\text{V} \quad A = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{5\text{V}}{50\text{mV}} = 100$$

$$A = 1 + \frac{R_2}{R_1} \Rightarrow \frac{R_2}{R_1} = 99 \text{ eg. } R_2 = 99\text{k}\Omega \quad R_1 = 1\text{k}\Omega$$

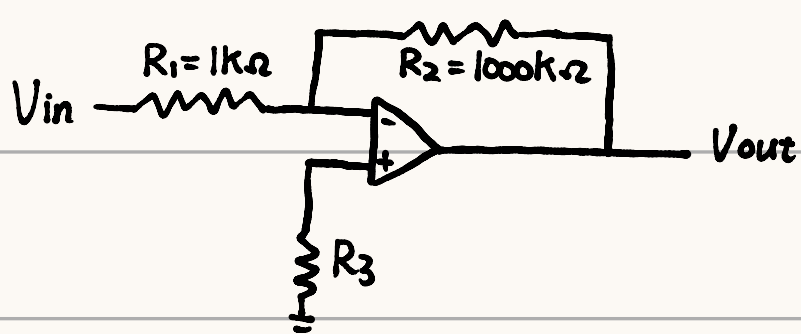


2. $V_{\text{rms}} = \sqrt{PR} = \sqrt{100 \times 50} \quad 50\sqrt{2}\text{V}$

$$V_{\text{max}} = 50\sqrt{2} \cdot \sqrt{2} = 100\text{V}$$

$$V_{\text{in max}} = 0.1\text{V}$$

$$A = -\frac{100}{0.1} = -1000 = -\frac{R_2}{R_1} \text{ eg. } R_2 = 1000\text{k}\Omega \quad R_1 = 1\text{k}\Omega$$



3. $V_{\text{out}} = -R_f \left[\frac{V_a}{R_a} + \frac{V_b}{R_b} + \frac{V_c}{R_c} + \frac{V_d}{R_d} \right]$

$$V_o = -60\text{k} \left[\frac{V_a}{R_a} + \frac{V_b}{R_b} + \frac{V_c}{R_c} + \frac{V_d}{R_d} \right] = -(3V_a + 5V_b + 4V_c + 2V_d)$$

$$\Rightarrow R_a = 20\text{k}\Omega \quad R_b = 12\text{k}\Omega \quad R_c = 15\text{k}\Omega \quad R_d = 30\text{k}\Omega$$

